Seminar 12 - Algebră

Baze. Dimensiuni

Fie V un K-sp vectorial (finit generat); $B \subseteq V$ bază $\dim V = |B|$
(dacă $\dim V = n$, orice n vector liniar independent din V formează o bază)

- $A, B \leq_K V$: $\dim(A+B) = \dim A + \dim B - \dim(A \cap B)$
- $A \leq_K V$: $\dim A = \dim V$: $A = V$

  $\dim A = 0$: $A = \{0\}$
- $f : V \to V'$ K-liniară : $\dim V = \underbrace{\dim \operatorname{Im} f}_{\text{rang}} + \underbrace{\dim \operatorname{Ker} f}_{\text{defect}}$

ex: Fie V, V' K-sp vect, $\dim V = \dim V' = n \in \mathbb{N}$

$f : V \to V'$ K-lin.

Ar că f inj $\Leftrightarrow$ f surj $\Leftrightarrow$ f izomorfism

Soluție: "$\Rightarrow$"

Pp f injectivă $\Leftrightarrow$ $\operatorname{Ker} f = \{0\}$

$\dim V' = \dim V = \dim \operatorname{Im} f + \underbrace{\dim \operatorname{Ker} f}_{=0}$

$\Rightarrow \dim \operatorname{Im} f = \dim V'$ (i)

Cum $\operatorname{Im} f \subseteq V' \overset{(i)}{\Rightarrow} \operatorname{Im} f = V' \Rightarrow f$ surj.

"$\Leftarrow$" Pp f surj $\Leftrightarrow \operatorname{Im} f = V'$

$\dim V = \underbrace{\dim \operatorname{Im} f}_{\dim V'} + \dim \operatorname{Ker} f \Rightarrow \dim \operatorname{Ker} f = 0$

$\dim V = \dim V'$

$\Rightarrow$ ~~dim~~ $\operatorname{Ker} f = \{0\}$

$\Rightarrow f$ inj

Obs:
$u = (u_1, u_2, u_3)$
$v = (v_1, v_2, v_3)$ (bază dacă sunt lin. indep)
$w = (w_1, w_2, w_3)$
$\in \mathbb{R}^3$

u, v, w formează o bază pt $\mathbb{R}^3 \Leftrightarrow$

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} \neq 0$$

• Fie $A \in \mathcal{M}_{mn}(K)$, $c_1, \dots, c_n \in K^m$ col. lui A

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ c_1 & c_2 & \dots & c_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

Nr de linii/col. lin indep din A = rang A

$\dim_K \langle c_1, \dots, c_n \rangle = \text{rang} A$

• În general, fie $x_1, \dots, x_n \in K^m$. Atunci $\dim_K$

$$\langle x_1, \dots, x_n \rangle = \text{rang} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ x_1 & x_2 & \dots & x_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \leftarrow \text{Componentele}$$

Dacă $x_1, \dots x_n \in V$, V- K-sp vect cu $\dim V = m$,

$B \subseteq V$ bază, $\exists f: V \to K^m$ izomorfism de sp vectorial a.î.

$f(\{e_1, \dots, e_m\}) = B \Rightarrow \dim_K \langle x_1, \dots, x_n \rangle = \text{rang} \begin{bmatrix} | & | & | & \dots \\ | & | & | & \end{bmatrix}$

coord lui v în baza B

ex: Fie $a = (-2, 1, 3)$

$b = (3, -2, -1)$

$c = (1, -1, 2)$

$d = (-5, 3, 4)$

$e = (-9, 5, 6) \in \mathbb{Q}^3$

Este adev că $\langle a, b \rangle = \langle c, d, e \rangle$?

generat de a, b

Sol: $\dim\langle a,b\rangle = \operatorname{rang}\begin{bmatrix} -2 & 3 \\ 1 & -2 \\ 3 & -1 \end{bmatrix} = 2$ (a, b lin indep)

$$\dim\langle c,d,e\rangle = \operatorname{rang}\begin{bmatrix} 1 & -5 & -9 \\ -1 & 3 & 5 \\ 2 & 4 & 10 \end{bmatrix} \underset{L_3-2L_1}{\overset{L_2+L_1}{=}}$$

$$= \operatorname{rang}\begin{bmatrix} 1 & -5 & -9 \\ 0 & -2 & -4 \\ 0 & 14 & 28 \end{bmatrix} = \operatorname{rang}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -4 \\ 0 & 14 & 28 \end{bmatrix} \overset{L_3(-\frac{1}{7})}{=}$$

$$= \operatorname{rang}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -4 \\ 0 & -2 & -4 \end{bmatrix} = \operatorname{rang}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 2 \Rightarrow$$ au aceeași dim

• $\langle a,b\rangle = \underbrace{\langle c,d\rangle}_{\shortparallel} \;?$

$\langle c,d,e\rangle$ doar că 2 din ele sunt lin ind

$\Leftrightarrow \langle a,b\rangle \subseteq \langle c,d\rangle \Leftrightarrow \begin{cases} a \in \langle c,d\rangle \\ b \in \langle c,d\rangle \end{cases}$

• $a \in \langle c,d\rangle \Leftrightarrow a, c, d$ lin dependenți $\Leftrightarrow \det = 0$

$\Leftrightarrow \begin{vmatrix} -2 & 1 & -5 \\ 1 & -1 & 3 \\ 3 & 2 & 4 \end{vmatrix} \overset{?}{=} 0$ „A" $(2C_1 - C_2 - C_3)$

• $b \in \langle c,d\rangle \Leftrightarrow b, c, d$ lin dependenți

$\Leftrightarrow \begin{vmatrix} 3 & 1 & -5 \\ -2 & -1 & 3 \\ -1 & 2 & 4 \end{vmatrix} = 0$

ex: Fie V K-sp vect, $\dim V = n \in \mathbb{N}^*$

$A, B \leq_K V$ a.î. $\dim A = n-1$ și $B \subseteq A$. Ar că:

• $A + B = V$ și $\dim(A \cap B) = \dim B - 1$

Sol: $\boxed{\dim(A+B) = \underbrace{\dim A}_{n-1} + \underbrace{\dim B - \dim(A\cap B)}_{\text{subpoțiu } \geq 1}}$

$B \not\subseteq A \Leftrightarrow A\cap B \subsetneq B \Rightarrow A\cap B \leq_K B$

dar nu coincide cu B

$\Rightarrow \dim(A\cap B) < \dim B$

$\Rightarrow \dim B - \dim(A\cap B) > 0$

$\Leftrightarrow \underbrace{\dim B - \dim(A\cap B)}_{\in \mathbb{N}} \geq 1$

$\dim(A+B) \geq n = \dim V \Rightarrow A+B = V$

$\dim(A\cap B) = \underbrace{\dim A}_{n-1} + \dim B - \underbrace{\dim(A+B)}_{=n}$

$= \dim B - 1$

ex: Fie V - K-sp vect, $A, B \leq_K V$ a.î. $\dim(A+B) = \dim(A\cap B)+1$

Ar că $A \subseteq B$ sau $B \subseteq A$

Sol: Pp $A \not\subseteq B$ și $B \not\subseteq A$. Ar că $\dim(A+B) \neq \dim(A\cap B)+1$

$\Leftrightarrow \dim(A+B) - \dim(A\cap B) \neq 1$

$B \not\subseteq A \Leftrightarrow A\cap B \subsetneq B \Rightarrow$

$\Rightarrow \dim(A\cap B) < \dim B$

$\Leftrightarrow \dim B - \dim(A\cap B) \geq 1 \quad (i)$

$A \not\subseteq B \Leftrightarrow B \subsetneq A+B$

$\Rightarrow \dim B < \dim(A+B)$

$\Leftrightarrow \dim(A+B) - \dim B \geq 1 \quad (ii)$

Din (i) și (ii) $\Rightarrow \dim(A+B) - \dim(A\cap B) \geq 2$

Am arătat $p \to q \Leftrightarrow \neg q \to \neg p$

ex: Fie V un K-sp vect, $f,g \in End_K(V)$ a.î. $f+g \in Aut_K(V)$, (ad. bij.)
$f \circ g = \theta$ (f,g transf lin din V în V)

Ar că $\dim V = \dim Imf + \dim Img$

Sol: $\dim V = \dim Imf + \dim Kerf$ (i)

$f \circ g = \theta \Leftrightarrow \forall x \in V : (f \circ g)(x) = 0$

$\Leftrightarrow \forall x \in V : f(g(x)) = 0$

$\Leftrightarrow \forall x \in V : g(x) \in Kerf$

$\Rightarrow Img \subseteq Kerf$

$\Rightarrow \dim Img \leq \dim Kerf$ (ii)

$\xrightarrow{i, ii} \dim V \geq \dim Imf + \dim Img$ (iii)

$f+g \in Aut_K(V) \Rightarrow f+g$ surj

$\Rightarrow V = Im(f+g) = \{f(x)+g(x) : x \in V\} \subseteq Imf + Img \subseteq V$

$\Rightarrow V = Imf + Img$

$\Rightarrow \dim V = \dim(Imf + Img)$

$= \dim Imf + \dim Img$

$= \dim(Imf \cap Img) \leq$

$\leq \dim Imf + \dim Img$ (iv)

$\xrightarrow{iii, iv}$ Concluzia

ex: Det baza și dimensiunea pentru $U, V, U+V, U \cap V$ ~~cand~~ cand:

a) $U = \langle u_1, u_2 \rangle$, $u_1 = (1,1,0,0)$, $u_2 = (1,0,1,1)$
$V = \langle v_1, v_2 \rangle$, $v_1 = (0,0,1,1)$, $v_2 = (0,1,1,0)$

b) $U = \langle u_1, u_2, u_3 \rangle$, $u_1 = (1,2,-1,2)$, $u_2 = (3,1,1,1)$, $u_3 = (-1,0,1,-1)$
$V = \langle v_1, v_2 \rangle$, $v_1 = (-1,2,-7,-3)$, $v_2 = (2,5,-6,-5)$

! orice generatori lin indep formează o bază

a) $\dim U = \text{rang}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix} = 2$ $\quad \{u_1, u_2\}$ bază pt U

$\dim V = \text{rang}\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} = 2$ $\quad \{v_1, v_2\}$ bază pt V

$\dim(U+V) =$

$= \dim \langle u_1, u_2, v_1, v_2 \rangle = \text{rang}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \dots = 4$

$\Rightarrow \{u_1, u_2, v_1, v_2\}$ bază pt $U + V = \mathbb{R}^4$

$\dim(U+V) = \dim U + \dim V - \dim(U \cap V) \Rightarrow$

$\Rightarrow \dim(U \cap V) = \dim U + \dim V - \dim(U+V) = 0$

$\Rightarrow U \cap V = \{0\}$ (tot bază)

b) $\dim U = \text{rang}\begin{bmatrix} \uparrow & \uparrow & \uparrow \\ u_1 & u_2 & u_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \dots = 3 \Rightarrow \{u_1, u_2, u_3\}$ bază pt

$\dim V = \text{rang}\begin{bmatrix} \uparrow & \uparrow \\ v_1 & v_2 \\ \downarrow & \downarrow \end{bmatrix} = \dots = 2 \Rightarrow \{v_1, v_2\}$ bază pt V

$\dim U + V = \dim \langle u_1, u_2, u_3, v_1, v_2 \rangle =$

$= \text{rang}\begin{bmatrix} 1 & 3 & -1 & -1 & 2 \\ 2 & 1 & 0 & 2 & 5 \\ -1 & 1 & 1 & -7 & -6 \\ -2 & 1 & -1 & -3 & -5 \end{bmatrix} = \dots = 3 \Rightarrow$

$\dim(U \cap V) = \dim U + \dim V - \dim(U+V) = 2 = \dim V$

$\Rightarrow V \subseteq U \cap V \Rightarrow V \leq U \cap V$